

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 23)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
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7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

A conservation organisation is converting 700 ha of arable farmland on the coast into a wetland reserve for breeding and wintering birds (as at Wallasea Island).

0 1

1 Explain why habitat area and shape are important in the design of the reserve. **[4 marks]**

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0 1

2 Explain how the reserve could be designed to provide a range of niches. **[3 marks]**

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0 1

3 Explain why some species may need to be deliberately introduced while others colonise naturally. **[2 marks]**

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0 1

4 Explain the value of a biological corridor linking the reserve to other wetlands. **[1 mark]**

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10

0 2 . 1

Explain how the development of biogeochemical cycles and of transpiration by land plants helped to maintain stable conditions for life.

[3 marks]

0 2 . 2

Describe **three** limitations of early methods of studying past conditions on Earth.

[3 marks]

1

2

3

0 2 . 3

Describe how each of the following modern techniques provides information about past conditions.

[4 marks]

Isotope analysis of ice cores

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Gas analysis of ice cores

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Radiocarbon dating

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Magnesium : calcium ratio in calcite

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10

0 3 Table 1 shows the ecological footprint and biocapacity per person for four countries and the world in 2016.

Table 1

	Ecological footprint / gha per person	Biocapacity / gha per person	Human Development Index
USA	8.1	3.6	0.92
UK	4.4	1.4	0.92
China	3.6	1.0	0.75
Bangladesh	0.8	0.4	0.61
World	2.8	1.7	0.72

0 3 . **1** Define **ecological footprint** and **biocapacity**.

[2 marks]

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0 3 . **2** Explain what is meant by a **global hectare** and why it is used.

[2 marks]

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03 . 3

Use Table 1 to calculate how many Earths would be needed if everyone lived like the average US citizen.

[1 mark]

Answer

Earths

03 . 4

Use Table 1 to describe the relationship between development and ecological footprint, and explain the concept of Earth Overshoot Day.

[3 marks]

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03 . 5

Suggest **two** ways a country such as the UK could reduce its footprint without reducing its HDI.

[2 marks]

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2

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10

0 4

Table 1 shows food conversion ratios (FCR) for livestock: the mass of feed needed to produce 1 kg of live mass gain.

Table 1

Livestock	FCR / kg feed per kg gain
Farmed salmon	1.2
Chicken	1.8
Pork	3.0
Beef (feedlot)	7.0
Sheep	8.0

0 4

1

Use Table 1 to explain why intensive poultry and fish production convert feed to food more efficiently than beef.

[3 marks]

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0 4

2

Explain **three** ways farmers reduce food chain energy losses in livestock production.

[3 marks]

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0 4

3

Explain why growing crops for direct human consumption usually produces more food per hectare than rearing livestock, and describe **one** situation where livestock is the better option.

[4 marks]

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10

05 . 1 Explain the effects of deforestation on soil.

[4 marks]

05 . 2 Explain how deforestation affects hydrology and climate.

[4 marks]

0 5 . 3

Give **two** ecological impacts of deforestation other than the direct loss of species.

[2 marks]

1

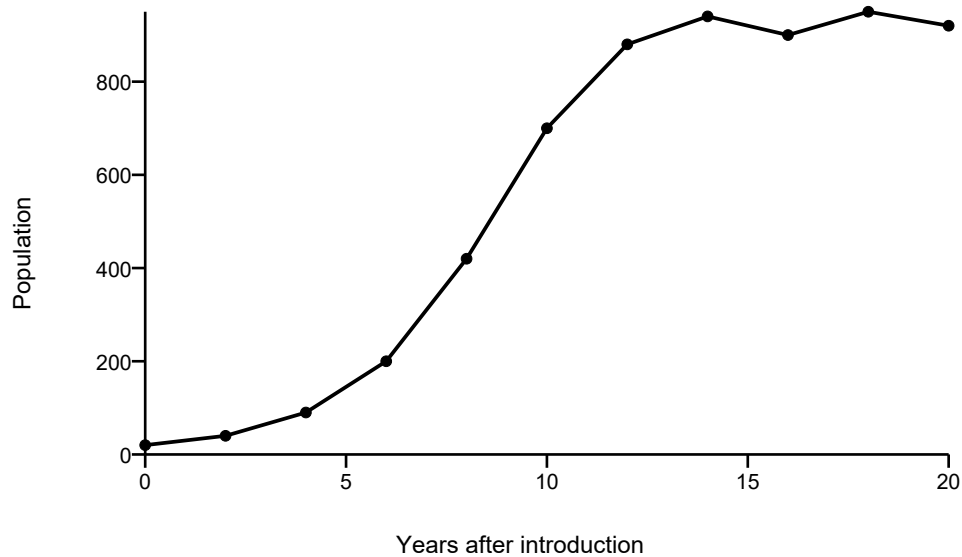
2

10

0 6

Figure 1 shows the population of red deer on an island after they were introduced in year 0. No predators are present.

Figure 1



0 6 . 1

Name the shape of the growth curve and explain the changes in growth rate shown.

[4 marks]

06 . 2

Explain the difference between density-dependent and density-independent factors, with an example of each.

[2 marks]

06 . 3

Explain why the deer population on the island may need to be culled even though it appears stable.

[2 marks]

06 . 4

Explain how predator and prey populations regulate each other.

[2 marks]

10

07 . 1

Explain why food surpluses produced in MEDCs in the 1970s could not easily be sold.

[3 marks]

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07 . 2

Describe **four** methods used to reduce over-production while maintaining farm incomes.

[4 marks]

1

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2

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07 . 3

Describe **three** features that farmers may be paid for under an agri-environmental scheme and the conservation purpose of each.

[3 marks]

1

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2

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3

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10

08 . 1

Explain why the ecosystem services provided by other species are often given a low conservation priority.

[2 marks]

08 . 2

Explain how soil organisms provide services that humans depend on.

[3 marks]

08 . 3

Explain why pollination by insects is more useful than wind pollination to plants growing in forests.

[2 marks]

08 . 4

Explain how seed dispersal by animals increases the survival of a plant's offspring.

[3 marks]

10

09 . 1

Which of the following is a feature of natural systems? Tick **one** box.

[1 mark]

A Reactions that require high temperatures

☐

B Reliance on non-renewable energy

☐

C Waste products that are the raw materials for other processes

☐

D Dispersal of elements so they cannot be re-used

☐

09 . 2

Explain why nitrogen fixation by bacteria is more sustainable than the Haber process.

[2 marks]

09 . 3

Explain why incineration of wastes is less sustainable than natural decomposition.

[2 marks]

5

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Describe the conservation efforts used to protect tropical rainforests and evaluate their effectiveness.

[9 marks]

Do not write outside the box

Extra space

10 . 2

State **one** reason why rainforest conservation is important to countries far from the tropics.

[1 mark]

10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.

[25 marks]

OR

$$\boxed{1} \boxed{1} . \boxed{2}$$

Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

END OF QUESTIONS